Representation Complexity of the SAT Equation

upper bounds, lower bounds, and hybrid compilation

Technical note complementary to *Compressed Queries on the SAT Equation*

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Abstract

The manuscript *Compressed Queries on the SAT Equation* separates the query layer from the compilation phase and proves that, once a representation of the forbidden region is built, queries are answered in time polynomial in the size of the representation. This note changes the question: instead of asking whether a universal representation solves $\mathsf{P} = \mathsf{NP}$, it studies which structural properties determine the minimum size of a representation and the cost of constructing it. We present four pieces: a landscape of the languages DSOP, ADSOP, ZDD, and Fourier with inclusions and unconditional separations; a region-composition theorem and a hybrid width profile; upper bounds parameterized by backdoors and treewidth; and a distinction between representation size, hybrid compilation size, and uniform compilation time. We also include a correction to the CEGAR loop: to detect a falsifying assignment outside the cover, one must query $\neg F \wedge \neg \mathrm{Cov}(\mathcal{C})$, not $F \wedge \neg \mathrm{Cov}(\mathcal{C})$. The note does not claim to prove $\mathsf{P} = \mathsf{NP}$; its goal is a map of bounds and an agenda of open problems.

Keywords: SAT Equation; knowledge compilation; lower bounds; upper bounds; DSOP; ADSOP; ZDD; Fourier; treewidth; backdoors.

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1 Introduction

The manuscript [1] proves that once a representation of the forbidden region $U(F)$ —DSOP, ADSOP, ZDD, or Fourier spectrum— is built, the queries bit, rank, select, #SAT, and sampling are answered in time polynomial in the size of the representation. The structural conclusion is that the bottleneck is not in the query: it is in the compilation. The same conclusion holds for the exact counter: the rank oracle is as hard as #SAT.

The question organizing this note is narrower than that of the original manuscript:

> Which structural properties of F determine the minimum cost of representing it and counting its models?

We do not ask whether the union of DSOP, ADSOP, ZDD, and Fourier solves P = NP. We ask which bounds can be proved for each language, what relations hold among them, and what separates the size of a representation from its compilation cost. This question allows unconditional results and explicit bounds even if universal compilation remains open.

The note has six contributions:

C1. a landscape of representations with inclusions and unconditional separations (Section 3);
C2. a region-composition theorem and a hybrid size $h_{\mathcal{R}}(F)$ (Section 4);
C3. a hybrid width profile and upper bounds parameterized by backdoors and treewidth (Section 5);
C4. an explicit separation between representation size, hybrid compilation size, and uniform compilation time (Section 6);
C5. a correction to the CEGAR loop and an analysis of verification complexity (Section 7);
C6. a list of open problems with falsifiable goals (Section 8).

The note does not assert P = NP or P ≠ NP. Its value, if any, lies in separating three types of results that are often confused: unconditional representation bounds, parameterized upper bounds, and conditional bounds under complexity hypotheses.

2 Preliminaries and notation

We use the notation of [1]. For a CNF F over n variables and m clauses, $N = n + m$ is the input size. $U(F) \subseteq \{0,1\}^n$ is the set of assignments that falsify F, and

$$\#\mathsf{SAT}(F) = 2^n - |U(F)|.$$

For a partial assignment $v \in \{0,1,\bullet\}^n$, $Q(v)$ is the subcube of assignments extending v, and $F|_v$ is the CNF obtained by fixing the variables of v. Thus,

$$U(F|_v) = Q(v) \cap U(F).$$

A *representation* of $F|_v$ is an exact description of $U(F|_v)$ or of $\mathsf{SAT}(F|_v)$ in one of the languages of the original manuscript: DSOP, ADSOP, ZDD, or Fourier spectrum. A *counter* for $F|_v$ is an algorithm that returns $\#\mathsf{SAT}(F|_v)$.

Representation	Parameter	Counting cost	Typical family
DSOP	K cubes	$O(K)$	wide clauses, few cubes
ADSOP	A affine pieces	$O(A)$	parity, affine subspaces
ZDD	Z nodes	$O(Z)$	bounded pathwidth/treewidth
Fourier	s coefficients	$O(s)$	parity, sparse spectrum
Counter	time T	T	treewidth, components, backdoor

Table 1: Representations, parameters, and counting cost. The rank and select costs are those of [1]. Representations may encode $U(F)$ or $\mathsf{SAT}(F)$; counting is adjusted by complement.

For a CNF F, $\mathrm{tw}(F)$ denotes the treewidth of the variable–clause incidence graph. A *backdoor set* $B \subseteq [n]$ fixes $b = |B|$ variables; for $\beta \in \{0,1\}^B$, $F|_\beta$ is the corresponding residual.

We distinguish three measures that should not be confused:

(i) the *representation size* $s_{\mathcal{L}}(F)$ in a fixed language $\mathcal{L}$;
(ii) the *hybrid compilation size*, which includes the tree and its leaves;
(iii) the *uniform compilation time*, the cost of an algorithm that constructs the representation and counts.

The third measure does not follow from the second. This distinction is developed in Section 6.

3 Representation landscape

3.1 Inclusions

The first relations among languages are immediate.

Proposition 3.1 (Basic inclusions)**.** *For every CNF F:*

(a) $A(F) \le K(F)$, because every cube is an affine subspace;
(b) $Z(F) \le O(nK(F))$, because a DSOP of K cubes is converted into a ZDD by building the trie of the K patterns;
(c) if a DSOP has cubes with supports $P_1, \dots, P_K$, then

$$s(F) \le \sum_{j=1}^{K} 2^{|P_j|}.$$

Proof. (a) is immediate. (b) The trie of K patterns of length n has at most nK nodes; the ZDD reduction can only decrease the size. The ZDD construction follows the decision-diagram representation of [9, 10]. (c) The Walsh–Hadamard transform is linear, and the indicator of a cube with support P has at most $2^{|P|}$ nonzero coefficients. Summing over the cubes gives the bound. □

These inclusions say that ADSOP and ZDD are at least as expressive as DSOP, and that Fourier can be large even when DSOP is small. The separations show that the inclusions are strict in concrete cases.

3.2 Unconditional separations

We use the Walsh–Hadamard transform in the form of [11].

Theorem 3.2 (Parity). *Let $O_n = \{x \in \{0,1\}^n : \bigoplus_{i=1}^n x_i = 1\}$. Every DSOP representing exactly O_n needs 2^{n-1} cubes. In contrast, O_n is a single affine subspace, its ZDD has $O(n)$ nodes, and its Fourier spectrum has two nonzero coefficients.*

Proof. A cube with at least one free coordinate contains two assignments that differ only in that coordinate; changing one bit changes the parity. Therefore a cube contained in O_n cannot have free coordinates: it is a singleton. Since O_n has 2^{n-1} points, a disjoint cover needs 2^{n-1} cubes. The affine representation is $x_1 \oplus \cdots \oplus x_n = 1$, of dimension $n-1$. The parity ZDD has two nodes per level, $O(n)$. The Fourier spectrum of the indicator of O_n is supported on $\varnothing$ and $[n]$. □

Theorem 3.3 (Wide clause). *Let C be a clause of width w. Then $U(C)$ is a single cube, so $K = A = 1$, and its ZDD has $O(n)$ nodes. Its Fourier spectrum has 2^w nonzero coefficients.*

Proof. The forbidden region of a clause is the cube that fixes its w literals; it is a cube and also an affine subspace. The ZDD of a cube has $O(n)$ nodes. The indicator of a cube with w fixed coordinates has spectrum supported on the 2^w subsets of those coordinates. □

Theorem 3.4 (Affine subspace and Fourier). *Let $H \subseteq \{0,1\}^n$ be an affine subspace of dimension d. Then H is a one-piece ADSOP and its Fourier spectrum has exactly 2^{n-d} nonzero coefficients. In particular, for $d = n/2$, ADSOP needs one piece and Fourier needs $2^{n/2}$ coefficients.*

Proof. The direction of H is a vector subspace V of dimension d. The Fourier transform of the indicator of H is supported on the orthogonal complement $V^\perp$, of dimension $n-d$; hence there are 2^{n-d} nonzero coefficients. The affine representation is a single piece. □

Table 2 summarizes the known values. Entries marked as open are problems from Section 8; they are not used in the bounds of this note.

Family	DSOP	ADSOP	ZDD	Fourier
Parity O_n	2^{n-1}	1	$O(n)$	2
Wide clause of width w	1	1	$O(n)$	2^w
Affine subspace of dimension d	open	1	open	2^{n-d}

Table 2: Separations between representation languages. Values are exact or proved bounds; in the affine-subspace row we consider a random family of dimension d, and "open" means that no lower bound is known for that family.

The landscape has a methodological consequence: a lower bound for DSOP is not a lower bound for SAT, for ADSOP, or for ZDD. The difficulty depends on the language. Accumulating separations is more informative than searching for a universal representation.

3.3 Parity, CNF, and encodings

Parity requires a precision. The set O_n is an affine subspace and has ADSOP 1. But an ordinary CNF encoding parity over the original variables needs 2^{n-1} clauses; a compact CNF with auxiliary variables has a different forbidden region $U(F)$, defined over all variables, and its ADSOP is not automatically one piece. Representation results must specify whether they refer to:

(i) the direct set O_n, without encoding;
(ii) a CNF with auxiliary variables, where $U(F)$ includes assignments that violate the encoding;

(iii) a mixed CNF–XOR formula, where parity constraints are native;
(iv) a projected counting problem, where one counts assignments to the original variables that have some auxiliary extension.

The distinction matters because a small representation of O_n does not necessarily translate into a small representation of the encoded CNF. In the rest of the note, when we speak of parity, we specify whether it is the direct set or an encoding.

A related problem is minimization: absolute DSOP minimization is NP-hard for functions given by truth tables [12]; the version with a CNF as input and K as a parameter remains open.

4 Region trees and composition

Definition 4.1 (Region tree)**.** A *region tree* for F is a binary tree T in which each internal node is labeled by a variable x_i and has two children, corresponding to $x_i = 0$ and $x_i = 1$. Each leaf ℓ is labeled by a partial assignment v_ℓ and by an exact representation R_ℓ of $U(F|_{v_\ell})$ or of $\mathsf{SAT}(F|_{v_\ell})$, or by an exact counter for $F|_{v_\ell}$. The partial assignments of the leaves are pairwise incompatible.

The incompatibility condition is what makes the leaves a partition of the cube: two leaves fix some variable with different values, so their cubes are disjoint, and every complete assignment ends at a leaf, so the union is the whole cube. In practice, the tree is built by branching on a free variable, and a leaf is not subdivided.

Theorem 4.2 (Composition by regions)**.** *Let T be a region tree for F with leaves $L(T)$. If each leaf ℓ allows $\#\mathsf{SAT}(F|_{v_\ell})$ to be counted exactly, then*

$$\#\mathsf{SAT}(F) = \sum_{\ell \in L(T)} \#\mathsf{SAT}(F|_{v_\ell}).$$

If the cost of counting leaf ℓ is t_ℓ, the total counting cost is $O(\sum_\ell t_\ell)$ plus the cost of constructing and traversing T.

Proof. The partial assignments of the leaves are pairwise incompatible, so the cubes $Q(v_\ell)$ are disjoint. Moreover, every complete assignment follows a unique path from the root to a leaf, so it belongs to one of those cubes; hence the union of the $Q(v_\ell)$ is $\{0,1\}^n$. Thus the sets $U(F|_{v_\ell}) = Q(v_\ell) \cap U(F)$ form a partition of $U(F)$, and the sets of assignments satisfying $F|_{v_\ell}$ form a partition of $\mathsf{SAT}(F)$. Adding the counts gives $\#\mathsf{SAT}(F)$. The cost is the sum of the leaf costs plus the traversal of the tree. □

Observation 4.3 (The theorem is not the difficulty)**.** Theorem 4.2 is the law of total counting in the guise of a tree. Its interest is not in the proof but in the question it leaves open: for which classes of CNFs does there exist a tree of polynomial size whose leaves are tractable, and can it be constructed in polynomial time? Without an answer to that question, the tree is an architecture, not a complexity result.

4.1 The hybrid size

To compare trees and representations, we define a size that includes the number of tree nodes and the size of its leaves.

Definition 4.4 (Hybrid size)**.** For a CNF F and a family of representations $\mathcal{R}$, let

$$h_{\mathcal{R}}(F) = \min_T \Big(|T| + \sum_{\ell \in L(T)} |R_\ell| \Big),$$

where T ranges over the region trees of F, $|T|$ is the number of nodes (each with unit cost), R_ℓ is the representation at leaf ℓ, and $|R_\ell|$ is its size (number of cubes, pieces, nodes, or coefficients). If a leaf is a counter, $|R_\ell|$ is the size of its description; its running time is accounted for separately in Theorem 4.2. When $\mathcal{R}$ is fixed, we write $h(F)$ for $h_{\mathcal{R}}(F)$.

The size $h_{\mathcal{R}}(F)$ generalizes the parameters of the original manuscript: if the tree has a single leaf, $h_{\mathcal{R}}(F)$ is of the order of the representation size K, A, Z, or s. If the tree branches down to single points, $h_{\mathcal{R}}(F)$ is of order 2^n. The question is whether some families attain the minimum with a small tree and leaves of different types.

Example 4.5 (Parity and a wide clause)**.** Let

$$F = (x_1 \oplus x_2 \oplus x_3 = 1) \wedge (x_4 \vee x_5).$$

The tree first branches on x_4. If $x_4 = 1$, the residual is the parity $x_1 \oplus x_2 \oplus x_3 = 1$, which is a one-piece ADSOP and has 8 models: 4 from the parity and 2 from x_5, which is free. If $x_4 = 0$, branch on x_5: if $x_5 = 1$, the residual is again the parity, with 4 models; if $x_5 = 0$, the residual is UNSAT, with 0 models. Hence

$$\#\mathsf{SAT}(F) = 8 + 4 + 0 = 12.$$

The tree has three leaves, all ADSOP or UNSAT. A DSOP of the satisfying region would have to combine the four parity points with the cubes of the clause, so the tree avoids that mixture. The example is small, but it shows the idea: the structure is chosen per region, not per formula.

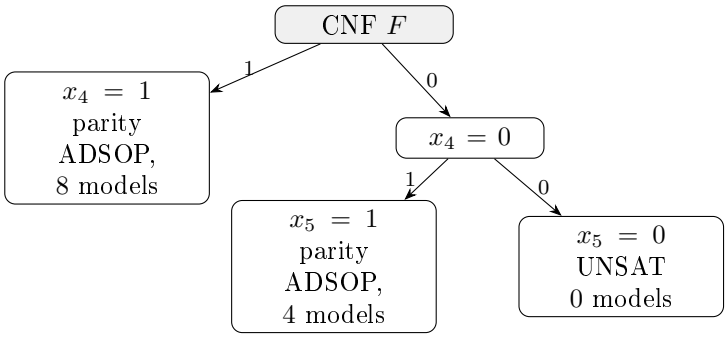


Figure 1: Region tree for Example 4.5. The leaves are disjoint and their counts add up.

5 Heterogeneous backdoors and upper bounds

The second piece is a local version of the tree: instead of branching to the end, one chooses a small set B of variables and solves the $2^{|B|}$ residuals with the appropriate representation. The notion of a heterogeneous backdoor is due to [2]; *backdoor trees* and the use of depth as a resource are studied in [4, 3]. What we add here is the combination with the representations of the original manuscript and an explicit treewidth bound.

Definition 5.1 (Heterogeneous backdoor)**.** Let F be a CNF and let $\mathcal{C}_1, \ldots, \mathcal{C}_m$ be classes of CNFs with an exact polynomial-time counter. A set $B \subseteq [n]$ is a *heterogeneous backdoor* for F if, for every $\beta \in \{0,1\}^B$, the residual $F|_\beta$ is unsatisfiable or belongs to some class $\mathcal{C}_j$; the class may depend on β.

Theorem 5.2 (Counting via a heterogeneous backdoor)**.** *If B is a heterogeneous backdoor of size b and every residual $F|_\beta$ is counted in time t_β, then*

$$\#\mathsf{SAT}(F) = \sum_{\beta \in \{0,1\}^B} \#\mathsf{SAT}(F|_\beta),$$

and the counting cost is $O(\sum_\beta t_\beta)$ plus the cost of finding B.

Proof. The assignments β are pairwise incompatible and cover all assignments to B. Therefore the residuals $F|_\beta$ partition the assignments of F, and their models add up. □

Example 5.3 (Affine backdoor)**.** For the formula of Example 4.5, $B = \{x_4, x_5\}$. The residuals are:

- $\beta = (1,0)$ and $\beta = (1,1)$: the parity $x_1 \oplus x_2 \oplus x_3 = 1$, with 4 models each;
- $\beta = (0,1)$: the same parity, with 4 models;
- $\beta = (0,0)$: UNSAT, with 0 models.

The total is 12. Every residual is affine or empty, so counting is linear algebra.

5.1 Treewidth upper bound

The backdoor becomes an explicit bound if every residual has low treewidth.

Proposition 5.4 (Treewidth bound for the residuals)**.** *Let B be a backdoor of size b. If every residual $F|_\beta$ has treewidth t_β, then exact counting is performed in time*

$$N^{O(1)} \sum_{\beta \in \{0,1\}^B} 2^{O(t_\beta)}$$

plus the cost of finding B and the tree decompositions. In particular, if $t_{\max} = \max_\beta t_\beta$,

$$T_{\#\mathsf{SAT}}(F) \leq 2^{b+O(t_{\max})} N^{O(1)}.$$

Proof. For each β, the standard dynamic program over a tree decomposition of the incidence graph of $F|_\beta$ counts the models in time $2^{O(t_\beta)} N^{O(1)}$. Summing over the 2^b residuals and adding the cost of constructing the decompositions gives the first bound. The second is the first with $t_\beta \leq t_{\max}$. □

The first expression is more informative than the second: it does not require all residuals to have the same complexity. Some may be very easy and others hard. This suggests defining a hybrid width profile:

$$\Phi(F) = \min_B \sum_{\beta \in \{0,1\}^B} 2^{c\, t_\beta},$$

where c is the constant of the treewidth counting algorithm. The profile is a structural bound; finding the set B that minimizes it need not be easy. This distinction between size and construction is the subject of Section 6.

5.2 Counting is not deciding

A decision backdoor is not necessarily a counting backdoor. Horn and 2-CNF formulas are tractable for SAT, but $\#2SAT$ is #P-complete [5]. Therefore, when defining a backdoor for counting, each residual must fall into a class that admits exact polynomial-time counting, not merely polynomial-time decision. This precision is essential for the upper bound of Theorem 5.2 to be valid.

The distinction also shows that the counting goal is stronger than the decision goal: a polynomial-time algorithm for #SAT would solve the complete counting problem and hence imply $\mathsf{P} = \mathsf{NP}$. This note does not seek that collapse; it seeks bounds for representations.

6 Size, compilation, and time

The knowledge-compilation literature distinguishes between the size of the final representation and the cost of constructing it. This distinction is central to interpreting the bounds of the previous sections.

Definition 6.1 (Three measures)**.** For a CNF F and a representation language $\mathcal{L}$:

(i) $s_{\mathcal{L}}(F)$ is the minimum size of an exact representation of F in $\mathcal{L}$;
(ii) $h_{\mathcal{R}}(F)$ is the minimum size of a region tree with leaves in $\mathcal{R}$, including the tree and the leaves;
(iii) $T_{\mathcal{L}}(F)$ is the time of a uniform algorithm that constructs the representation in $\mathcal{L}$ and counts.

The measures satisfy $s_{\mathcal{L}}(F) \leq h_{\mathcal{R}}(F)$ when $\mathcal{L} \in \mathcal{R}$ and the tree has a single leaf; here $s_{\mathcal{L}}$ denotes the size in the language and should not be confused with the Fourier sparsity $s(F)$. But $T_{\mathcal{L}}(F)$ is not bounded by $s_{\mathcal{L}}(F)$. A small representation may be hard to find; an upward compilation procedure may require exponential time or space even when the final representation has constant size. The knowledge-compilation literature studies this phenomenon [7, 8].

There is a second trap: if a leaf may be an arbitrary exact counter, a lower bound over all choices of representations would amount to a lower bound for exact counting itself. For a bound to be informative, one must restrict the leaf classes or parameterize their cost. In this note, the leaves are DSOP, ADSOP, ZDD, Fourier, or counters with a declared cost t_β.

6.1 A conditional check with #ETH

Under the counting variant of the Exponential Time Hypothesis (#ETH), formulated analogously to ETH [6], there is no uniform algorithm that solves $\#3SAT$ in time $2^{o(n)}$. Therefore, if a uniform hybrid compiler —tree construction, leaf compilation, and counting— solved all instances of $\#3SAT$ in time $2^{o(n)}$, it would contradict #ETH.

This observation is not a new lower bound for the hybrid parameter: it is a consistency check. To obtain a new lower bound, one would have to prove that the construction of the tree or some leaf must be large for a concrete family. The observation fixes the framework: lower bounds must be stated for families of inputs and uniform algorithms, not for an isolated formula whose result could be precomputed.

7 Verification and certificates

The third piece is a refinement loop that constructs and verifies the representation at the same time. It is an adaptation of CEGAR/CEGIS to the language of regions.

For brevity, we write $\mathrm{Cov}(\mathcal{C})$ for the union of the regions of a cover $\mathcal{C}$.

Algorithm 1 `CegarCompile`(F)

Require: CNF F
Ensure: an exact verified cover of $U(F)$, or UNAVAILABLE

$\mathcal{C} \leftarrow \emptyset$
while true do
  propose a region R (cube or affine subspace) contained in $U(F)$ and disjoint from $\mathcal{C}$
  if none exists, **break**
  check $F \wedge R$: if SAT, refine R (maintaining disjointness from $\mathcal{C}$) and continue
  add R to $\mathcal{C}$
  check $\neg F \wedge \neg \mathrm{Cov}(\mathcal{C})$: if SAT, propose a region around the witness, disjoint from $\mathcal{C}$
end while
verify that $\mathrm{Cov}(\mathcal{C}) = U(F)$; if not, return UNAVAILABLE
return $\mathcal{C}$

The formula $\neg F$ is encoded with selectors or Tseitin, and affine regions are encoded as XOR equations; the solver can accept them natively or translate them to CNF. In both cases the query is a SAT instance of polynomial size in N.

Proposition 7.1 (Correctness of the loop)**.** *If* `CegarCompile` *terminates with a verification that* $\mathrm{Cov}(\mathcal{C}) = U(F)$*, every added region satisfies* $R \subseteq U(F)$*, and every region is disjoint from the previous ones, then* $\mathcal{C}$ *is a disjoint and exact cover of* $U(F)$*, and* $\#\mathsf{SAT}(F) = 2^n - \sum_{R \in \mathcal{C}} \mathrm{vol}(R)$*.*

Proof. The check $F \wedge R$ UNSAT proves $R \subseteq U(F)$. The check $\neg F \wedge \neg \mathrm{Cov}(\mathcal{C})$ UNSAT proves that there is no falsifying assignment outside the cover. Therefore the union is exactly $U(F)$. Since every region is added disjoint from the previous ones, the volumes add up. The counting identity is the definition of $\#\mathsf{SAT}$. □

Observation 7.2 (The correct coverage query)**.** The coverage query must be $\neg F \wedge \neg \mathrm{Cov}(\mathcal{C})$, not $F \wedge \neg \mathrm{Cov}(\mathcal{C})$. The second searches for assignments satisfying F outside the cover; the first searches for assignments falsifying F outside the cover, which is exactly what must be detected to prove $\mathrm{Cov}(\mathcal{C}) = U(F)$. An earlier version of this note used the incorrect query; the error is corrected here.

7.1 Complexity of verification

Theorem 7.3 (Containment of a cube)**.** *Given a CNF* F *and a cube* R*, deciding whether* $R \subseteq U(F)$ *is coNP-complete.*

Proof. Membership in coNP is clear: a witness that $R \not\subseteq U(F)$ is an assignment $x \in R$ that satisfies F, and it is verified in polynomial time. For hardness, given a CNF G, take $F = G$ and $R = Q_{\mathrm{all}} = \{0,1\}^n$. Then $R \subseteq U(G)$ if and only if every assignment falsifies some clause of G, that is, if and only if G is unsatisfiable. Hence the problem is coNP-hard. □

Corollary 7.4 (DSOP verification)**.** *Verifying that a list of cubes* $\mathcal{C}$ *is an exact DSOP of* $U(F)$ *is coNP-complete. With a resolution trace (or a trace from another polynomial-time verifiable proof system), verification is polynomial in the size of the trace.*

Proof. Verification includes checking $R \subseteq U(F)$ for every $R \in \mathcal{C}$, which is coNP-hard by Theorem 7.3; it also includes coverage $U(F) \subseteq \mathrm{Cov}(\mathcal{C})$, checked by $\neg F \wedge \neg \mathrm{Cov}(\mathcal{C})$. The disjointness of two cubes is checked in polynomial time by testing whether they fix some variable with different values. The problem is in coNP: a failure witness is an assignment satisfying F inside some R, or an assignment falsifying F outside the cover; both are verified in polynomial time. With traces, each containment is verified in polynomial time in the trace. □

Observation 7.5 (The certificate)**.** Each region needs a proof that $F \wedge R$ is UNSAT; total coverage needs a proof that $\neg F \wedge \neg \operatorname{Cov}(\mathcal{C})$ is UNSAT. Without a trace, verifying the containment of a cube in a union of cubes is coNP-hard, as Theorem 7.3 shows. With a trace, the certificate is verifiable in polynomial time in the trace. The loop does not promise a short trace; it promises that, if the loop terminates, the certificate is verifiable.

Observation 7.6 (The cost)**.** Each iteration of the loop makes at least one SAT query. The number of iterations can be exponential; the algorithm is correct, but it is not a polynomial compiler. Its value is in engineering and certification: it allows the hybrid engine to be built with verifiable guarantees and detects when a proposed representation is too large.

8 Open problems

Open Problem (Lower bound for ADSOP)**.** Prove or refute that there is a family of CNFs with $m = \operatorname{poly}(n)$ whose minimum ADSOP has size $2^{\Omega(n)}$. Parity does not work: it is the case where ADSOP is one piece. One must look for a family with high affine complexity, for example a random function or a union of affine subspaces with controlled intersections.

Open Problem (ZDD for affine subspaces)**.** Prove that a random affine subspace of dimension $n/2$ has ZDD $2^{\Omega(n)}$ for every variable order, or find an order with polynomial ZDD. An affirmative answer would separate ADSOP from ZDD: ADSOP 1 versus exponential ZDD. The natural technique is an OBDD lower bound for linear codes via communication complexity.

Open Problem (Fourier versus ADSOP)**.** Find a function with a polynomial-size Fourier spectrum and exponential ADSOP. The opposite direction is already proved: an affine subspace of dimension $n/2$ has ADSOP 1 and Fourier $2^{n/2}$. The other direction would decide whether Fourier and ADSOP are incomparable.

Open Problem (Hybrid profile)**.** Find families of CNFs where $\Phi(F)$ is provably smaller than $2^{O(\operatorname{tw}(F))}$ or than the cost of compiling a single representation (DSOP, ADSOP, ZDD, or Fourier), and find an algorithm that constructs a backdoor B attaining a bound close to $\Phi(F)$. Without the latter, $\Phi(F)$ is a structural measure, not an algorithm.

Open Problem (Size versus compilation)**.** Prove that there is a family of CNFs with a polynomial-size final representation but uniform compilation time $2^{\Omega(n)}$, or prove that for a rich class small size implies polynomial compilation. The distinction between $s_{\mathcal{L}}(F)$ and $T_{\mathcal{L}}(F)$ is the central point.

Open Problem (Parameterized complexity of compilation)**.** Determine the complexity of finding a heterogeneous backdoor of size b and of finding tree decompositions of width t_β for the residuals. The upper bound of Proposition 5.4 assumes that B and the decompositions are available; the cost of finding them is part of the problem.

Open Problem (Short certificates)**.** Can the `CegarCompile` loop produce polynomial-size certificates for a rich class of CNFs? Without short traces, coverage verification remains the bottleneck.

9 Conclusions

The original manuscript proves that compressed querying is possible once a representation is available. This note shifts attention to the complexity of representations: which languages are more compact, which separations can be proved, which upper bounds are obtained by conditioning on a backdoor, and what distance separates the size of a representation from the cost of constructing it.

The unconditional results are the inclusions and separations of Section 3, the composition by regions, the correction of the CEGAR loop, and the coNP characterization of verification. The upper bounds of Section 5 are parametric: they depend on the size of the backdoor and on the treewidth of the residuals. The main lesson is that the size of a representation does not determine the compilation cost, and that a lower bound for one language does not transfer to another.

There is no proof of $\mathsf{P} = \mathsf{NP}$ here. There is a map of bounds and an agenda of open problems: ADSOP, ZDD for affine subspaces, Fourier versus ADSOP, the hybrid profile, and the gap between size and compilation. The next step is not to add more speculative routes but to close one of these bounds with a theorem. The criterion of success is not that the tree combines known methods, but that it yields a new bound not obtained by applying each method separately.

A Two technical precisions

A.1 The two overlapping cubes

An earlier version of this note used the formula

$$F = (x_1 \vee x_2) \wedge (x_3 \vee x_4)$$

and wrote

$$U(F) = Q(00\bullet\bullet) \cup Q(\bullet\bullet 00).$$

The equality is correct, but the two cubes overlap at 0000, so they do not form a DSOP. A disjoint cover is

$$U(F) = Q(00\bullet\bullet) \,\dot\cup\, Q(1\bullet 00) \,\dot\cup\, \{0100\}.$$

The volumes are $4+2+1=7$. Two disjoint cubes cannot cover the set because 7 is not the sum of two powers of two; the minimum DSOP has three terms. The example shows that a compact cover may fail to be disjoint and that disjointization has a real cost.

A.2 Fourier normalization

The original manuscript uses the unnormalized Walsh–Hadamard transform,

$$\hat f(S) = \sum_{x\in\{0,1\}^n} f(x)(-1)^{\sum_{i\in S} x_i},$$

so the coefficient of a cube $Q(u)$ with fixed-coordinate set $P = \{i : u_i \neq \bullet\}$ is

$$\widehat{1_{Q(u)}}(S) = \begin{cases} 2^{n-|P|}\prod_{i\in S}(-1)^{u_i}, & S \subseteq P, \\ 0, & \text{otherwise.} \end{cases}$$

If one uses the normalized convention $\hat f(S) = 2^{-n}\sum_x f(x)(-1)^{S\cdot x}$, the factor is $2^{-|P|}$. In both cases P must be defined explicitly; it should not be left implicit in $\mathrm{supp}(u)$ without clarifying that u is the ternary pattern of the cube.

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